

7 - Assignment on theoretical data transformations

Problem 1

Movies rated	Mov 1	Mov 2	Mov 3	Movie 4	Movie 5	Movie 6	Movie 7	Movie 8
User u_1	5	1	4	.	.	.	2	1
User u_2	5	.	3	2	5	.	2	1
User u_3	1	4	2	5	2	.	5	4
User u_4	4	2	.	1	5	2	3	3

Use collaborative filtering to predict User 2's preference for Movie 6.

Problem 2

Use Pearson correlation to determine which movie is the most similar to Movie 1.

(Optional: Google how to use a for loop to speed up this task.)

Problem 3A

Apply mean imputation grouped according to region to fill in missing values in the column for "Agriculture" and the column for "Industry" for the countries dataset from

<http://manctsui.github.io/datasets/countries.csv>

Problem 3B

How many NA's did you have to fill from each region?